CHAPTER 3

# MOTION IN A PLANE

**CHAPTER ESSENCE AT A GLANCE:**
Motion in a plane is just TWO independent straight-line motions happening simultaneously! Horizontal speed never changes because there is zero horizontal gravity. Vertical speed rises and falls like a thrown ball. Combine them and you get the famous parabola!

## 4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Scalar vs Vector:** Scalars have only magnitude (mass, speed, work). Vectors have magnitude AND direction (velocity, acceleration, force, momentum).

• **Parallelogram Law:** Resultant of two vectors A and B at angle θ: R = sqrt($A^2$ + $B^2$ + 2AB cos θ). Direction: tan α = (B sin θ) divided by (A + B cos θ). If |A + B| = |A - B| implies θ = 90° (Perpendicular!).

• **Projectile Independence:** Horizontal motion has zero acceleration (v x = u cos θ = const). Vertical motion is free fall under gravity (a y = -g). At maximum height, vertical speed v y = 0, but horizontal speed remains u cos θ!

• **Complementary Angles:** Horizontal range R is identical for complementary launch angles θ and (90° - θ). E.g., launch at 30° and 60° gives the exact same range! Maximum range R max = $u^2$ divided by g occurs at θ = 45°.

## ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}, \quad \tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$

*Parallelogram Law Resultant and Direction*

$$T = \frac{2u\sin\theta}{g}, \quad H = \frac{u^2\sin^2\theta}{2g}$$

*Time of Flight and Maximum Height of Projectile*

$$R = \frac{u^2\sin 2\theta}{g} \quad \left(R_{\max} = \frac{u^2}{g} \text{ at } \theta = 45^\circ\right)$$

*Horizontal Range of Projectile*

$$a_c = \frac{v^2}{r} = \omega^2 r$$

*Centripetal Acceleration in Uniform Circular Motion*

## CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Range is equal for complementary angles (θ and 90° - θ). E.g., launch at 30° and 60° gives the EXACT same horizontal range!

• **Alert:** At the highest point of projectile motion, the speed is NOT zero! It is still moving horizontally at u cos θ!

## HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

**1. State the Parallelogram Law of vector addition. [Score: 2 Marks]**

**Answer & Key Valuation Steps:**

• **[Step 1: Statement]** If two vectors acting simultaneously at a point are represented in magnitude and direction by two adjacent sides of a parallelogram drawn from a common vertex, their resultant is represented in magnitude and direction by the diagonal of the parallelogram passing through that common vertex. [2 Marks]

• **Key Exam Result:** R = sqrt($A^2$ + $B^2$ + 2AB cos θ) along diagonal

**2. Show that the horizontal range of a projectile is the same for two angles of projection θ and (90° - θ). [Score: 2 Marks]**

**Answer & Key Valuation Steps:**

• **[Step 1: Range at Angle θ]** R(θ) = ($u^2$ sin 2 θ) divided by g. [1 Mark]

• **[Step 2: Range at Angle 90 - θ]** R(90 - θ) = [$u^2$ sin 2(90 - θ)] divided by g = [$u^2$ sin(180 - 2 θ)] divided by g. Since sin(180 - x) = sin x: R(90 - θ) = ($u^2$ sin 2 θ) divided by g = R(θ). [1 Mark]

• **Key Exam Result:** R(θ) = R(90° - θ) (Equal for complementary angles)

**3. Derive expressions for the magnitude R and direction α of the resultant of two vectors A and B inclined at an angle θ. [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• **[Step 1: Vector Geometry]** Let vectors OP = A and OS = B be inclined at angle θ. Complete parallelogram OPQS. Drop perpendicular QN to OP produced. In right triangle QNP: PN = B cos θ, QN = B sin θ. [1 Mark]

• **[Step 2: Resultant Magnitude]** In right triangle ONQ: $OQ^2$ = $ON^2$ + $QN^2$ = $(OP + PN)^2$ + $QN^2$ = $(A + B \cos \theta)^2$ + $(B \sin \theta)^2$ = $A^2$ + 2AB cos θ + $B^2$ ($\cos^2 \theta + \sin^2 \theta$) = $A^2$ + $B^2$ + 2AB cos θ. R = √($A^2$ + $B^2$ + 2AB cos θ). [1 Mark]

• **[Step 3: Direction α]** tan α = QN divided by ON = (B sin θ) divided by (A + B cos θ). [1 Mark]

• **Key Exam Result:** R = sqrt($A^2$ + $B^2$ + 2AB cos θ), tan α = (B sin θ) divided by (A + B cos θ)

**4. Define uniform circular motion. Derive an expression for centripetal acceleration a c = $v^2$ divided by r = $\omega^2$ r. [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• **[Step 1: Definition]** When a particle moves along a circular path of radius r with constant speed v, its motion is called uniform circular motion. Its velocity direction changes continuously. [1 Mark]

• **[Step 2: Velocity Change Geometry]** In time Δt, angular displacement is Δθ. Arc length Δs = r Δθ. Position vectors r1, r2 and velocity vectors v1, v2 form similar isosceles triangles. |Δv| divided by v = |Δr| divided by r => |Δv| = (v divided by r) |Δr|. [1 Mark]

• **[Step 3: Centripetal Acceleration]** a c = lim(Δt → 0) [|Δv| divided by Δt] = (v divided by r) lim(Δt → 0) [|Δr| divided by Δt] = (v divided by r) v = $v^2$ divided by r. Since v = r ω: a c = $(r \omega)^2$ divided by r = $\omega^2$ r (directed towards center). [1 Mark]

• **Key Exam Result:** a c = $v^2$ divided by r = $\omega^2$ r (radially inward)

**5. A projectile has a range of 50 m when projected at an angle of 15°. What will be its horizontal range when projected at an angle of 45° with the same initial speed? [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• **[Step 1: Range Formula at 15°]** R = ($u^2$ sin 2 θ) divided by g. For θ = 15°: 50 = [$u^2$ sin(2 × 15°)] divided by g = [$u^2$ sin 30°] divided by g = ($u^2$ divided by g) x half (0.5). [1.5 Marks]

• **[Step 2: Calculate $u^2$ divided by g]** $u^2$ divided by g = 50 × 2 = 100 m. [0.5 Mark]

• **[Step 3: Range at 45°]** For θ = 45°: R 45 = [$u^2$ sin(2 × 45°)] divided by g = [$u^2$ sin 90°] divided by g = ($u^2$ divided by g) x 1 = 100 m. [1 Mark]

• **Key Exam Result:** Horizontal Range at 45° = 100 m

**6. Given two vectors A = 2 i + 3 j + 4 k and B = 4 i - 2 j + 3 k, calculate: (a) The scalar product A · B (b) The angle θ between vectors A and B. [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• **[Step 1: Dot Product]** A · B = (2)(4) + (3)(-2) + (4)(3) = 8 - 6 + 12 = 14. [1 Mark]

• **[Step 2: Magnitudes]** $|A| = \sqrt{(2^2 + 3^2 + 4^2)} = \sqrt{(4 + 9 + 16)} = \sqrt{(29)}$. $|B| = \sqrt{(4^2 + (-2)^2 + 3^2)} = \sqrt{(16 + 4 + 9)} = \sqrt{(29)}$. [1 Mark]

• **[Step 3: Angle θ]** cos θ = (A · B) divided by (|A| |B|) = 14 divided by ($\sqrt{(29)}$ x $\sqrt{(29)}$) = 14 divided by 29 = 0.4828. $\theta = \cos^{-1}$(14 divided by 29) ≈ 61.1°. [1 Mark]

• **Key Exam Result:** A · B = 14, $\theta = \cos^{-1}$(14 divided by 29) ≈ 61.1°